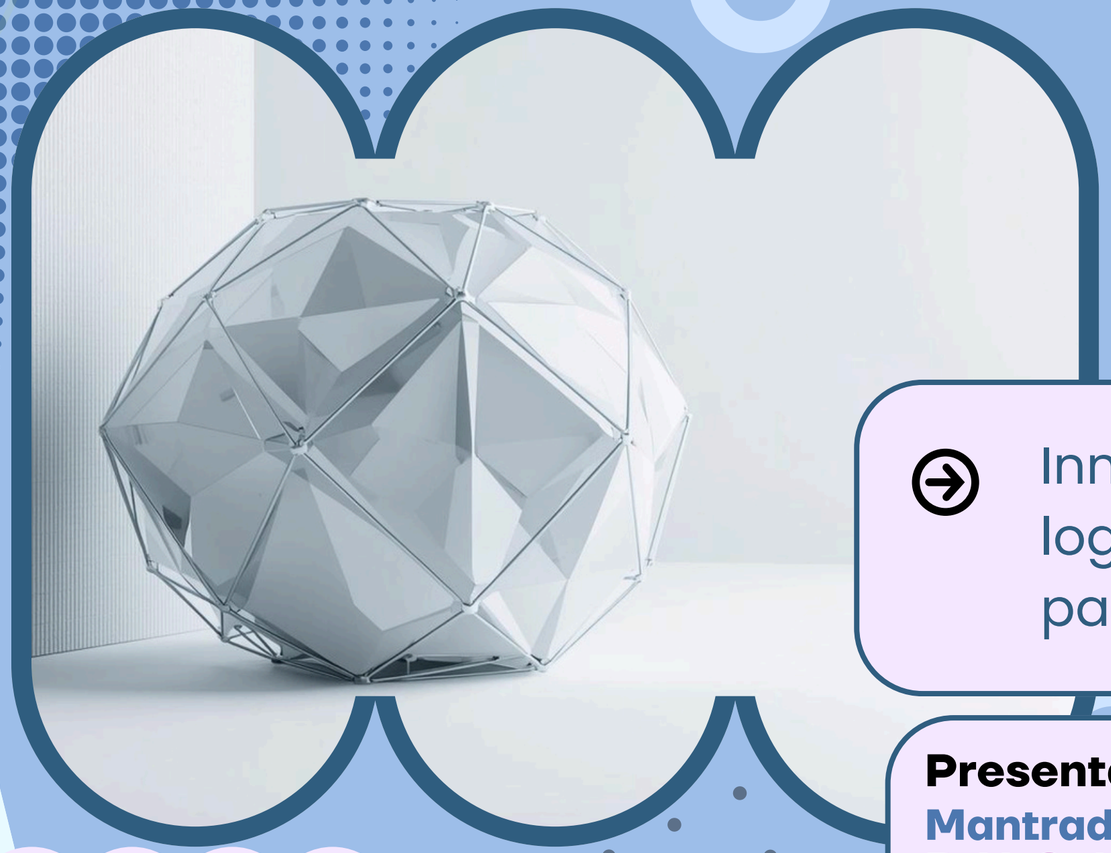
NeuraX 2026 Hackathon

Morpho-Pack – Generative Geometric Sustainability



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Innovative solution for irregular logistics and sustainable packaging

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Introduction to Problem Statement

- **The Conflict:** Global e-commerce relies on standardized cuboid boxes for irregular products (e.g., cycles, agricultural tools).
- **The Waste:** Shipping "air" leads to Volumetric Deficiency, where up to 50% of a container’s volume is empty space filled with non-recyclable "Void-Fill" (plastic peanuts/air pillows).
- **The Environmental Cost:** Inefficient stacking causes a "Carbon Ripple"—more trucks on the road, higher "Last-Mile" emissions, and massive material wastage.
- **The Breakage Gap:** Poorly secured irregular items lead to high return rates, doubling the carbon footprint of the product lifecycle





Literature Survey

Here are the latest Research papers and reports of 2025-2026:

Title	Authors	Publication / Journal	Year	Key Features for Software Implementation
Packing approach to generate void structures in lightweighted parts	Chernov, Y., Stoyan, Y., & Pankratov, O.	*Mathematics* (MDPI)	2026	**Geometric Logic:** Uses **Phi-functions** to define the "No-fit Polygon" (NFP) logic. In software, this translates to the algorithm that prevents our "shrinkwrap" from intersection of products.
Optimization method for green packaging recycling path based on Bat Algorithm	Li, H., & Wang, J.	*Journal of Dynamics and Games* (AIMS)	2026	**Carbon Calculation:** Provides the metaheuristic formula for the **"Carbon Ripple."**
Sustainable Packaging 2025: Top barriers for packaging purchasers	McKinsey & Company	*Industry Insights Report*	2025	**UI/UX Strategy:** Identifies the need for **"Digital Integration.** This justifies use of **Azure** making high-end custom packaging accessible to non-experts via a low-code **Power App**.
Biodegradable Packaging Market: Global Industry Analysis 2026–2036	Future Market Insights	*Market Intelligence Report*	2026	**Database/Lifecycle Logic:** Provides the degradation rates for Mycelium, PHA, and Bio-resins allowing prediction of material "breakdown" in different environments.

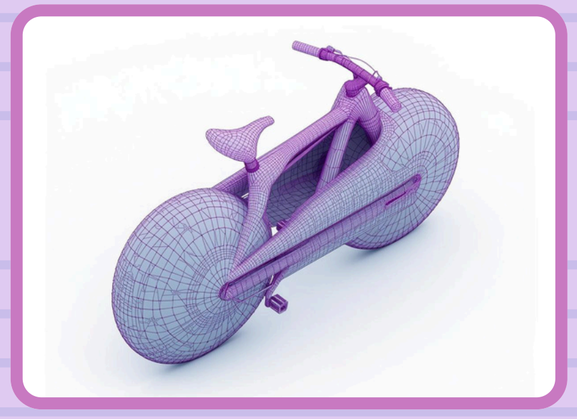
Literature Survey

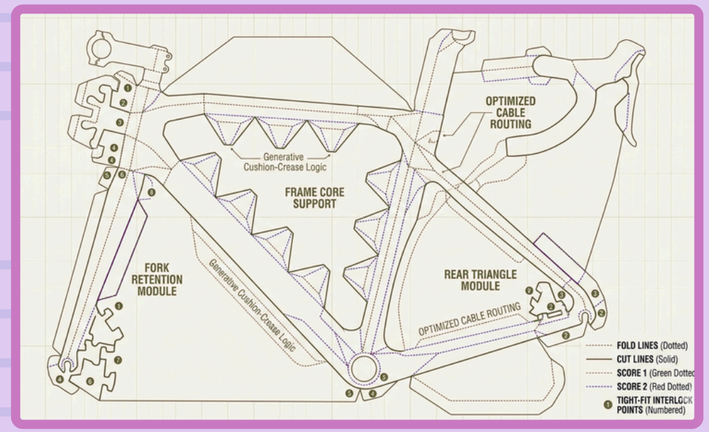
Technology Gap Analysis Table

Research Area	Key Findings (2024-2026)	The "Morpho-Pack" Gap
Generative AI in Packaging	Research shows Gen-AI can reduce design cycles by 70% and material thickness by 15% via structural optimization.	Most Gen-AI tools currently focus on *graphics/branding*, not *generative 3D geometry* for irregular shapes.
Life Cycle Assessment (LCA)	Latest LCAs confirm that "Bio-PE" and "Mycelium" have significantly lower carbon footprints, but only if "End-of-Life" (EoL) pathways are clear.	LCA data is rarely integrated *into* the design tool; it's usually an afterthought or a separate audit.
Computational Geometry	Recent advances in "Nesting Algorithms" (2D irregular packing) show that optimizing cutting patterns can save up to 25% of raw sheet stock.	These algorithms are rarely applied to *3D-to-2D* unfolds for custom, tight-fitting product shells.

Solution Overview

Morpho-Pack –Generative Geometry and Sustainability





Automates Production:

Generates a ready-to-cut 2D Flat Pattern that maximizes material yield and minimizes off-cut waste.

Shrinkwraps Reality

Uses generative algorithms to create a tight-fitting structural "net" around irregular 3D objects (like a bicycle).

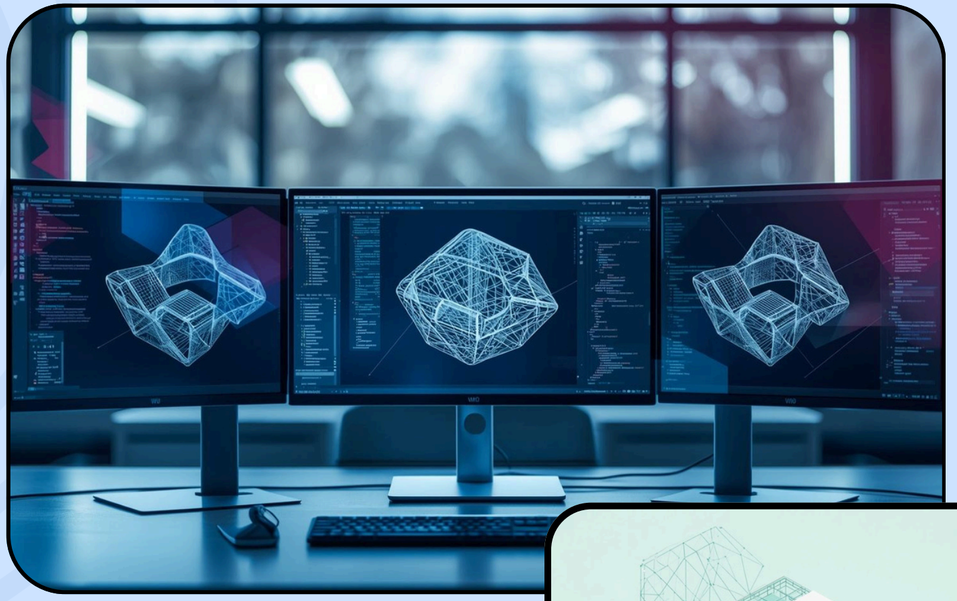
Eliminates Plastic

Replaces void-fill with "Cushion-Creases"—geometry-based cardboard supports generated automatically.

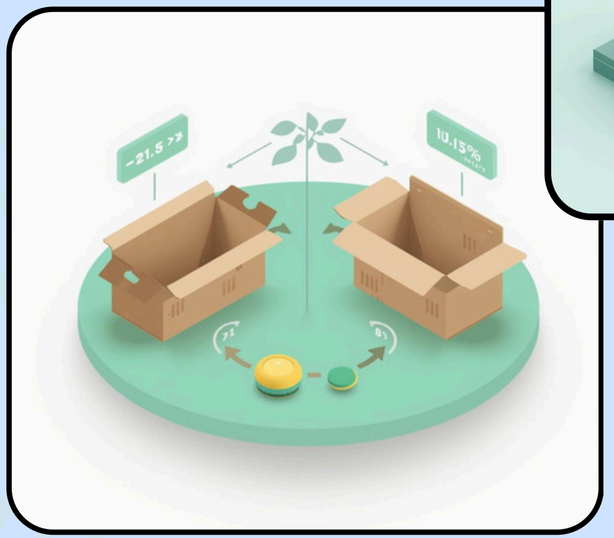
Calculates the Ripple:

Provides a real-time Integrated Sustainability Score (S_s) based on volume, material circularity, and carbon savings.

Tech Stack







3D Geometry Engine

Blender (Python API) & Fusion 360.

Generative AI

Generative AI: Azure OpenAI (GPT-4o). Used to analyze mesh vertices and suggest optimal "Seam Placement" for the 2D unfold.

Automation & Flow

Microsoft Power Automate. Triggers headless Blender scripts upon file upload.

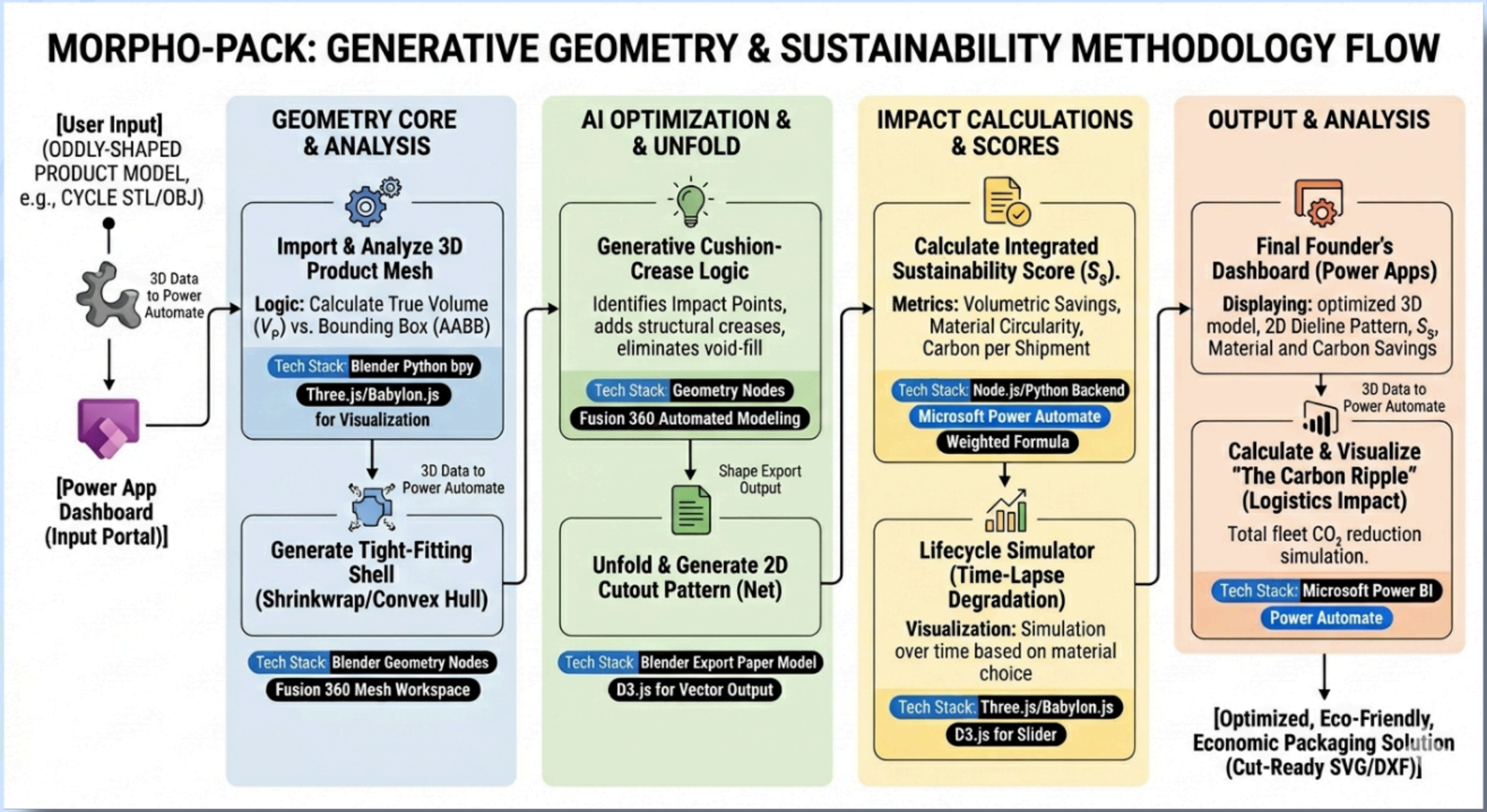
Logic & Analytics

Azure Functions (Serverless) for heavy math; Power BI for the Carbon Ripple dashboard.

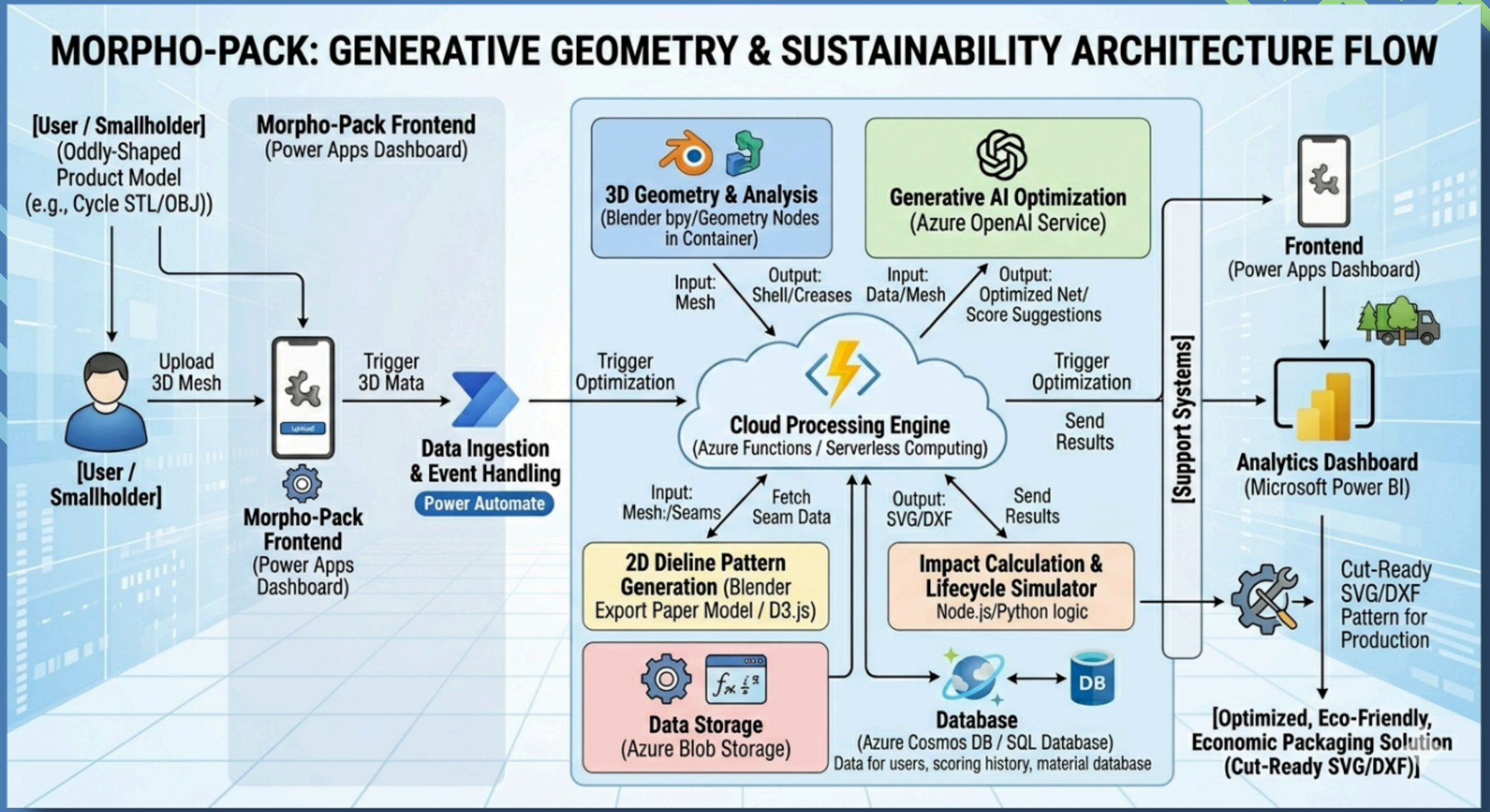
Frontend

Microsoft Power Apps for a low-code, cross-platform founder interface.

Methodology



System Architecture

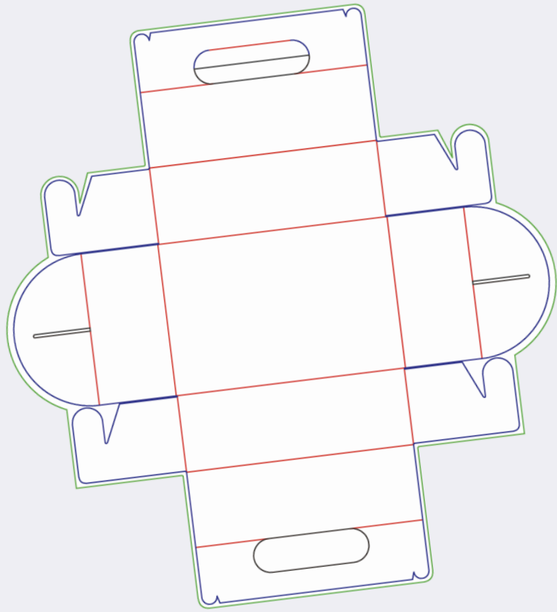


Novelty

Why "Morpho-Pack" is Different

- **Zero-Void Design:** We don't find a box; we grow a box. It moves from "Boxing" to "Morphing."
- **Material-Awareness:** The solution integrates a Lifecycle Simulator—visualizing how the package degrades in real-time based on material choice (e.g., Mushroom packaging vs. Cardboard).
- **Democratized Customization:** Bringing high-end engineering (Generative Design) to local farmers and small businesses via a simple mobile interface.
- **Pro-Human Friction**: A UI that highlights the "Ethical Choice," showing the user the impact of their material selection on SDG targets.

Future Scope



- **AR-Guided Assembly:** Using Microsoft HoloLens / Mobile AR to show warehouse workers exactly how to fold the complex 2D cutout around the irregular product.
- **Edge Computing:** Deploying the optimization logic to Azure IoT Edge for real-time packaging on factory floors.



- **Blockchain Provenance:** Integrating a ledger to track the Circularity Score of the package as it moves through the supply chain and back into recycling.
- **Smart-Fiber Integration**: Designing patterns for "active" packaging that monitors the ripening of agricultural products (linking to your Crop Recommendation interest).

Thank You